Breez

BiPAP Non-Invasive Ventilator with Sensor Fusion & BLE Telemetry

ROB-GY 6103 (Advanced Mechatronics) · NYU Tandon · Raj Priyadarshi, Devank Thawre, Sagarika Rao Valluri

Abstract

Breez is a proof-of-concept non-invasive ventilator built on BiPAP (Bilevel Positive Airway Pressure) principles for post-COVID respiratory rehabilitation and sleep-apnea support. Air is delivered from an Ambu bag by a stepper-driven pump while a suite of biomedical sensors reads the patient's vitals directly from the breath.

A MAX30102 pulse oximeter captures heart rate and blood oxygen (SpO■), and a ScioSense ENS160 air-quality sensor tracks the CO■, NO■, and O■ content of the exhaled air, alongside dual piezoresistive pressure sensors — all sharing one I²C bus. Two Arduino UNOs split the work master/slave, and every reading streams over an RN-42 Bluetooth link to a companion Android app.

Technical Highlights

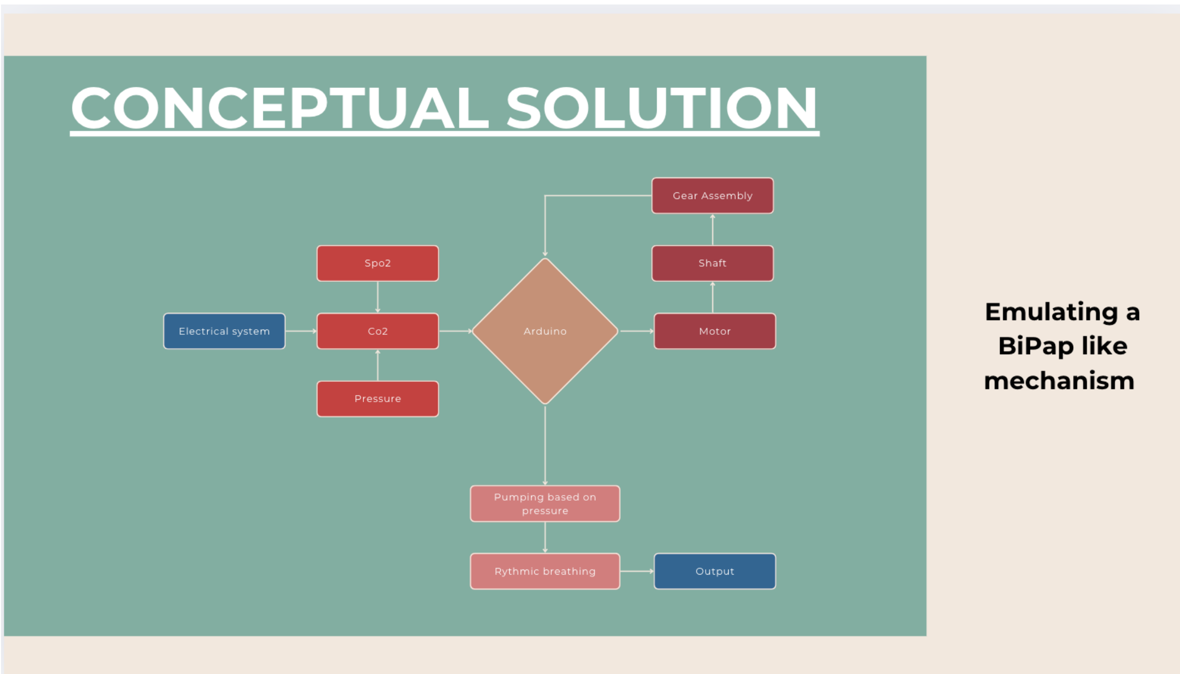
- **Sensor fusion from the breath:** heart rate and SpO■ from the MAX30102, and CO■ / NO■ / O■ from the ENS160 air-quality sensor, combined on a single I²C bus.
- **Airway pressure sensing:** dual piezoresistive pressure sensors read through a 24-bit HX711 ADC for high-resolution nasal/airway pressure.
- **Distributed control:** two Arduino UNOs operate master/slave — one handles sensor fusion, the other drives the stepper pump.
- **Live wireless telemetry:** an RN-42 Bluetooth link streams heart rate, SpO■, and gas readings to a companion Android app that scans, connects, and displays live status.
- **Stepper-actuated pump:** an A4988 driver microsteps a stepper motor driving a geared Ambu-bag compression mechanism.

Technologies Used

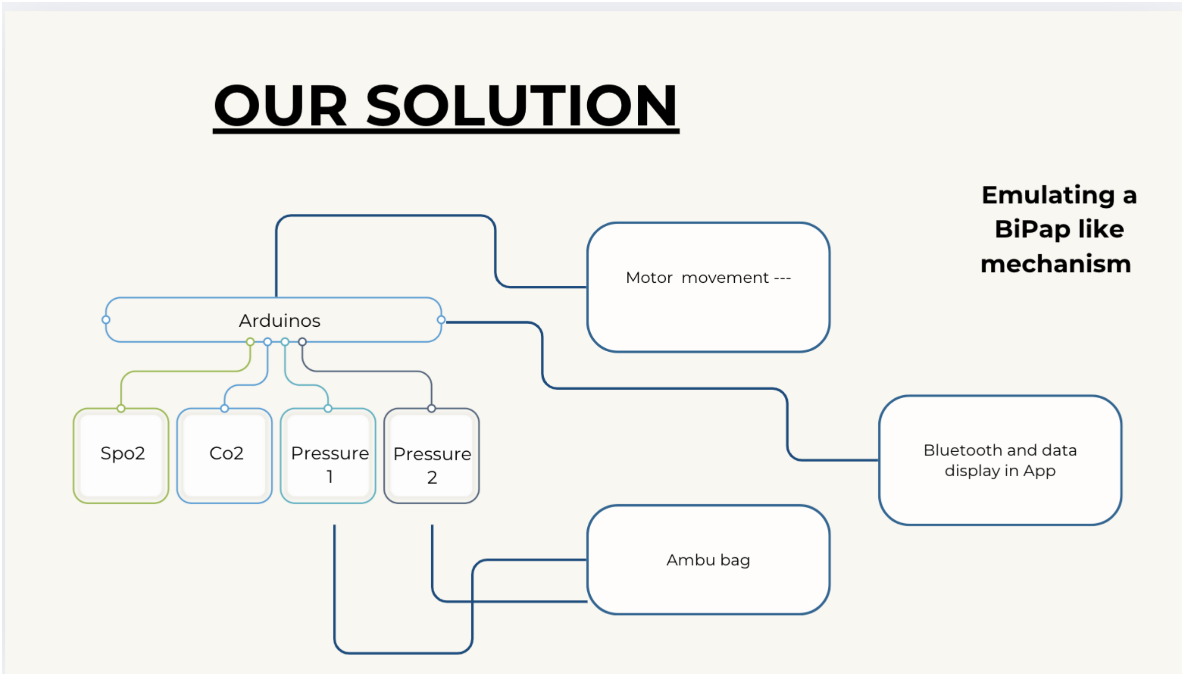
- **Languages:** C++ (Arduino), Java (Android)
- **Controllers:** Dual Arduino UNO (I²C master/slave)

- **Sensors:** MAX30102 (SpO■ / heart rate), ScioSense ENS160 (CO■ / NO■ / O■), piezoresistive pressure + HX711 24-bit ADC
- **Buses / protocols:** I²C, SPI, UART / SoftwareSerial
- **Wireless:** RN-42 Bluetooth telemetry to Android app
- **Actuation:** Stepper motor + A4988 driver, geared Ambu-bag pump

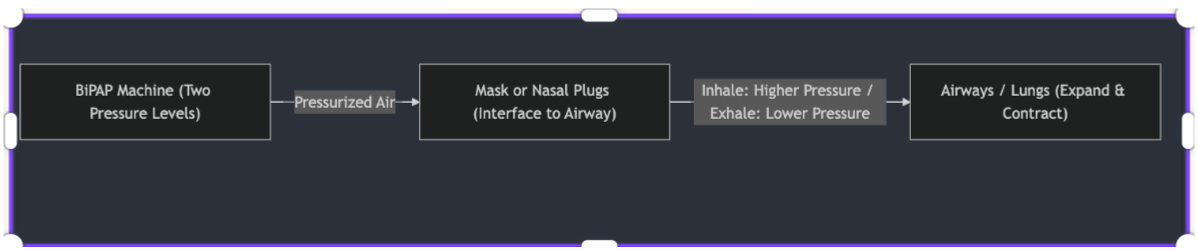
Figures



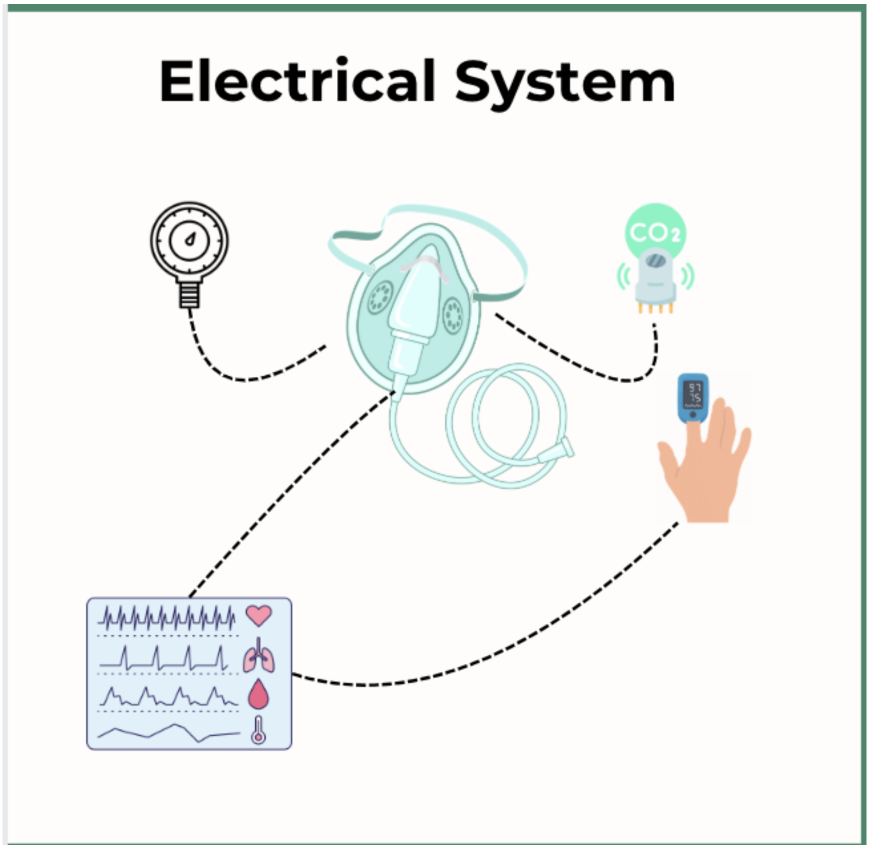
Conceptual solution — the sensing-to-actuation flow.



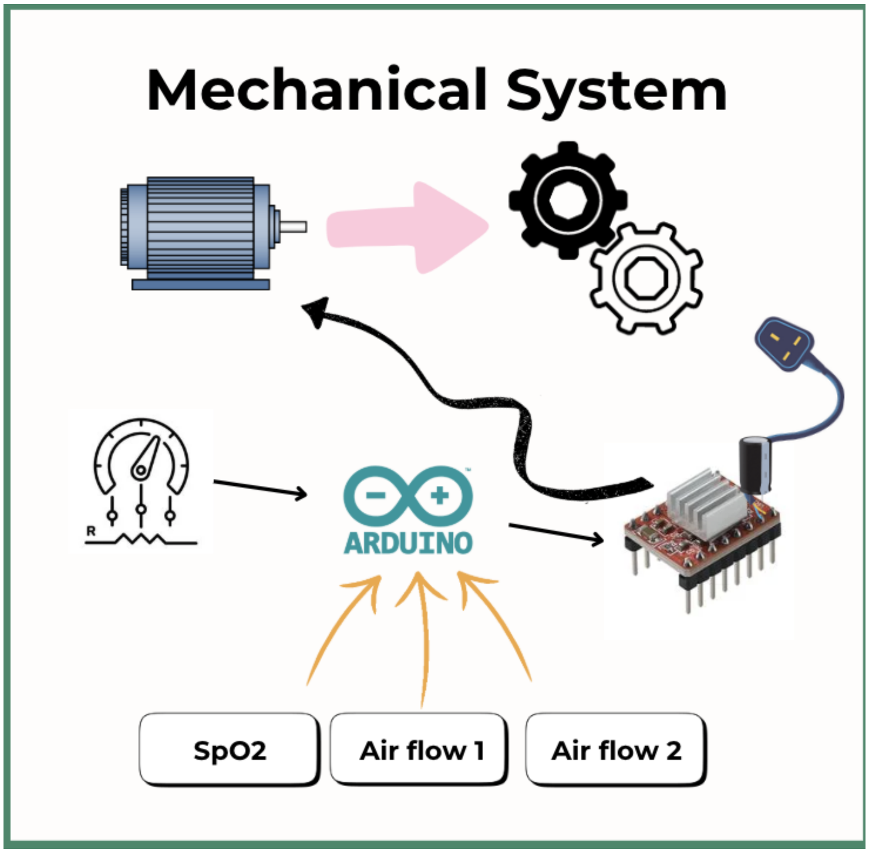
System block diagram — sensors, the two Arduinos, pump, and BLE link.



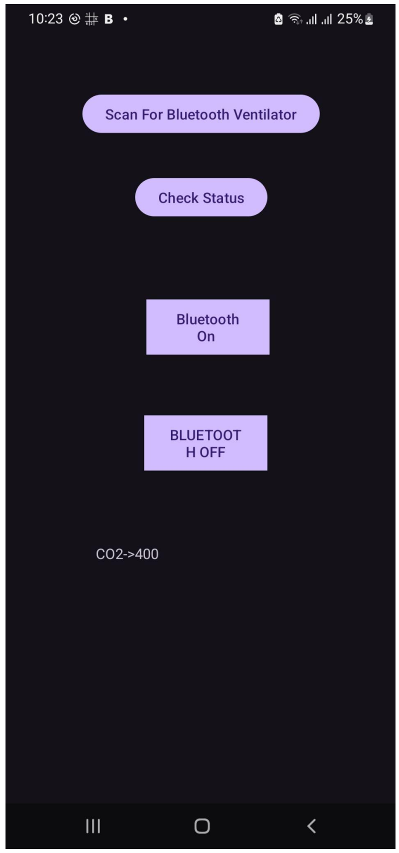
BiPAP airflow — bilevel positive airway pressure from the Ambu bag through the mask.



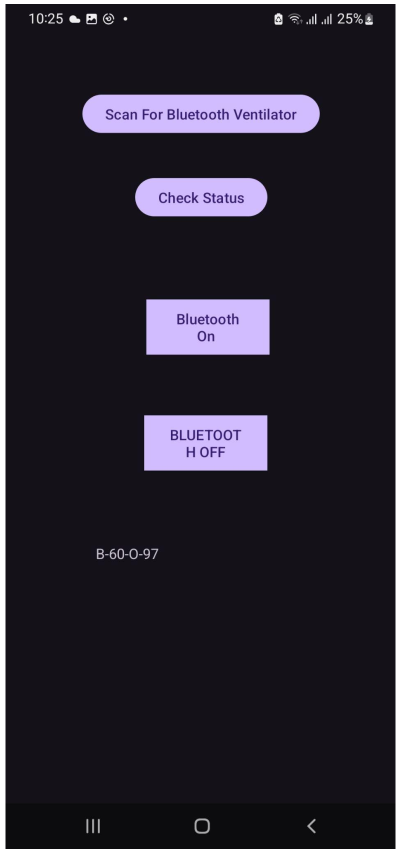
Electrical system — I²C sensor bus, HX711 pressure front-end, stepper driver, Bluetooth.



Mechanical system — geared stepper-driven Ambu-bag compression mechanism.



Companion Android app scanning for and connecting to the ventilator.



Live telemetry — CO■, SpO■, and heart rate over Bluetooth.

Outcome

Breez demonstrates a complete telemedical ventilator concept: multi-sensor vitals captured directly from the breath, fused on a shared I²C bus, coordinated across two microcontrollers, and streamed live to a mobile app — a compact platform for remotely-monitored respiratory support.